Engine Model(s): PT6A

77-20-01

T5 INDICATING SYSTEM - DESCRIPTION AND OPERATION

1. Description and Operation

The interturbine temperature (T5) system consists of the bus-bar, probes, wiring harness, T1 trim thermocouple , and related parts.

The T5 bus-bar and probe assembly is located on the outer section of the power turbine stator housing with the probes passing through ports in the housing. The probes are individually fastened to their respective bosses with coupling nuts, while the bus-bar is bolted to the rear flange of the stator housing.

The T5 wiring harness provides the electrical interconnection between the bus-bar and probes and the electrical wiring for airframe instrumentation via the harness terminal block mounted on the gas generator case. The alumel and chromel terminal lugs at the inner ends of the twin-lead harness are connected to their respective terminals on the bus-bar assembly. The leads are welded to internal connections within the terminal block.

The T1 trim thermocouple is located at the 2 o'clock position immediately above the air inlet screen on the gas generator case. It is attached by an integral flange to the center fireseal at its front end, and supported by a loop clamp at the rear. The leads at the front of the thermocouple are connected to the T5 terminal block on the gas generator case.

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